DHAN SCMS

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Executive Summary

The System that we will use in Supply Chain Management Systems or (SCMS) will be a web-based solution to meet the needs of MEGAION, this system aims to automate and innovate inventory management by implementing scannable barcodes with location tracker for every machine. The system has a real-time status dashboard with separate tabs for inventory and location tracker.

The importance of maintaining accurate records makes the supply chain management system essential. Our team is developing the DHAN Supply Chain Management System (SCMS) to address these challenges, providing a solution for MEGAION to efficiently store.

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I. Introduction

In the contemporary medical industry, reliable supply chain management is critical for ensuring the timely availability of essential medical equipment (MRI, Electrolytes analyzer, hospital beds, non-consumables). The complexity of handling various medical equipment, the need for fast and efficient tagging, and the imperative to maintain accurate records necessitate a comprehensive supply chain management system. Recognizing these challenges, our team will develop the DHAN Supply Chain Management System (SCMS), a solution specifically designed for MEGAION, involved in storing, distributing, and delivering medical equipment.

1.1 Project Context

For businesses that distribute and store medical equipment, such as MEGAION in Paranaque City, an efficient supply chain management system is essential for success. The nature of medical supplies demands precise inventory tracking, procurement efficiency, logistics coordination, and strict adherence to regulatory guidelines. Traditional supply chain management systems, often reliant on manual processes, are susceptible to human errors and delays. These shortcomings can jeopardize the availability of vital equipment, highlighting the urgent need for a more advanced, reliable, and user-friendly supply chain management system.

1.2 Statement of the Problem

Our client, a medical equipment distributor, is struggling with their inventory management due to a large manual process. This makes it hard to maintain accurate and up-to-date records. The complexity of their operations and strict regulatory requirements further complicates the situation.

The specific problems they are facing includes:

1. **Manual process of tracking stock levels for delivery or demonstration of machines, leading to potential delays**. MEGAION recognizes the need for a reliable inventory management system to mitigate these risks. According to Edson Maybituin, the Inventory Manager, he struggles to keep up with the current system and the extensive manual process, citing increased error susceptibility.

2. **There are instances when the products are understocked, leading to supply disruptions.** According to Edson Maybituin, the inventory manager, he also stated that there is a problem with an advance purchase order with currently zero (0) stock for machine equipment.

3. **Outdated and inefficient process of machine retrieval after demonstration.** According to Edson Maybituin, the Inventory Manager, machines used for demonstrations are not being retrieved. There are times when these machines are not logged into the system, resulting in missing stock.

1.3 Objectives

To automate and streamline the inventory management system of the medical equipment distributor using barcodes, ensuring accurate tracking, real-time status updates, and integrate supply chain management system.

To resolve these challenges, the client aims to automate their inventory system, and here are the specific objectives:

1. To reduce the delay in checking stock levels for items that are for delivery or demonstration by 95%.
2. **To optimize processes and ensure seamless coordination among stakeholders** in the supply chain of MEGAION, particularly when sourcing hospital equipment from various suppliers.
3. To streamline the process for retrieving and logging demonstration machines to mitigate instances of missing stock.

1.4 Significance of the Project

Fixing the client's problem is the main goal of creating this project. Adapting to modern technologies and incorporating them into the client's business will enhance the current system that they are using. Furthermore, creating this system will negate any employees' hindrances in doing their work. The project will benefit the following:

- Business Owner/s:

  This project will benefit the business owner/s because they will have an improved system that will assist their challenges in encoding items, profiling data, and checking their own inventory. Furthermore, the client could utilize various project features that will help manage their business.

- Service Employees:

  This project will benefit the service employees by providing them with a way to manage what goes in and out of their inventory, what available stocks they currently have, and the items being prepared to sell. This will help minimize errors for any employees using the system.

- Developers

    As for the developers, this project will benefit them because it gives them broader knowledge on how to tackle a problem and think of a technological solution that comes along with it. The developers would have a clear understanding of how to plan the project, build it, and implement it to solve the problem they're facing.

Sustainable Development Goals:

SDG No.9: Industry, Innovation and Infrastructure

- To promote innovation in modern technologies, facilitating local and international trade in the medical industry.

1.5 Scope and Limitations

The supply chain management system will automate the inventory management, procurement, and logistics coordination. The focus of this project will be on items that go in and out of the inventory, procurement efficiency, and logistics. The focus will be implementing important features such as building user dashboard and modules. Furthermore, the creation, selection, and deletion of products will also be part of its features. Moreover, the system will also cover reporting that will provide a centralized dashboard with real-time data and customizable reports. The barcode scanning functionality of the system will automatically demonstrate if a product is for delivery or for demonstration. It will also be automatically added to the inventory dashboard of the system. If the product is for demonstration, there will be a notification system that will notify the client in retrieving the machines after demonstration. In procurement management, a feature would be added in placing orders for products and making invoices. A real-time location tracker will also be added to the system. This will help in tracking the location of machine equipment that is for demonstration. Lastly, there will be an AI assistant that will be used in demand forecasting, inventory needs, and potential supply chain disruptions.

This project will only focus on inventory management, procurement, and logistics. Other matters such as company expenditures and sales will not be involved in the project. Lastly, the project will only limit its coverage to the current developers and clients that will be able to access the product.

II. Review of Related Literature / Systems

2.1 Related Applications

Anaplan

A software called Anaplan is well-known for its sophisticated analytics features, which include machine learning and predictive analytics to improve forecast accuracy. The software also facilitates scenario planning, which enables companies to assess the possible effects of different situations. Because of Anaplan's platform's great degree of customization, companies may easily adapt their forecasting models to meet their unique requirements. Because of this versatility, businesses can include specific workflows, business rules, and scenarios into their forecasting procedures. High data volumes and complicated models can be handled by Anaplan without affecting performance. Large organizations or companies with intricate supply chains that need to handle and analyze big datasets to produce precise forecasts must have this scalability. [1]

Magaya

For handling the entire logistics process—from quotation to billing—Magaya is the finest option. With Magaya, users may quickly and accurately provide quotes for clients based on current pricing, which include carrier rates, surcharges, and other costs. This feature makes quoting easier and aids in more effectively securing business. Users can also handle all shipment kinds on a single platform, including air, sea, and ground. In addition to creating shipments, combining cargo, and handling the associated paperwork, this also entails giving a comprehensive progress report for each shipment. The software comes with features for managing compliance with customs laws, such as creating the required paperwork and electronically submitting it to the appropriate customs authorities. [1]

Netstock

Netstock's ability to foresee demand is one of its advantages. It helps businesses manage ideal stock levels and lower the risk of stockouts or excess inventory by using sophisticated algorithms and past sales data to forecast future inventory demands. Advanced analytics tools in the program offer information on inventory performance. For proactive inventory control, it can draw attention to problems including overstock, understock, and product obsolescence. Multiple departments and outside partners are frequently involved in effective SCM. All parties involved in inventory management may easily communicate and coordinate thanks to the collaboration options offered by Netstock.[1]

Ordoro

Ordoro is an online platform that helps you manage all aspects of your shipping. An online tool called Ordoro assists you in handling every facet of your shipment. With this platform, you can manage returns from one or more locations, assess inventory, print postage, and give consumers tracking information. Ordoro is compatible with a number of e-commerce sites and marketplaces, such as Shopify, Etsy, Amazon, and eBay. From a single dashboard, you'll be able to simply process shipping labels for each order and keep track of them all [2].

Cin7

Cin7 is a cloud-based inventory management tool made to assist companies incontrolling their stock, orders, and sales via a variety of distribution methods. It is a complete solution for companies trying to optimize their operations since it incorporates multiple features like order management, inventory control, point of sale (POS), and warehouse management [3].

Odoo

Odoo is an all-in-one program for managing businesses. Numerous applications are available for it, including inventory management, HR administration, sales and purchase management, accounting, and much more. It is simple to use for both administrators and end users thanks to its intuitive UI. Apart from the typical functionalities of accounting software, Odoo comes with a range of pre-built business processes that can be tailored to your specific requirements or utilized exactly as is [4].

2.2 Related Literature

2.2.1 Marketing Adaptation to Modern Technologies

Marketers must stay ahead of the technological curve due to the applications and solutions that follow technological advancements, which have never-before-seen consequences and results. For instance, it's critical to determine how new technology will probably affect business strategy as well as consumer (and customer) behaviors. In addition, new developments in data storage, analytics, and technology present marketing professionals with chances to generate, share, collect, and provide worth for their clientele [5].

2.2.2 Implementation of Automated Inventory System to Medium-sized Enterprises

The advantages and disadvantages of setting up an automated inventory management system for SMEs are thoroughly discussed in the case study of the business. The business was able to reduce costs, simplify its inventory management processes, and improve client support after implementing the new technology. The case study demonstrates how important it is for small and medium-sized businesses to consider the advantages and disadvantages of utilizing an automated mechanism for managing inventories. Prior to starting such projects, SMEs ought to assess the technical complexity of the procedure for implementation, the requirement for employee training, and the possible return on investment [6].

2.2.3 Implementation of Barcode on Warehouse Management Systems for efficiency

Barcode implementation in warehouses has many benefits, including the ability to reduce errors in receiving goods and speed up this process; can automatically determine the location of storage; can minimize errors in storing goods at the storage area; can reduce errors in location and goods picked by the picker; it can help identify excess or insufficient inventory; it can determine the suitability and quality of the goods to be shipped; it reduces human error in checking goods; it uses less paper; and it speeds up the process of issuing manifests for shipping [7].

2.2.4 Optimizing SCM with Python

Pandas and scikit-learn, two of Python's data analysis tools, enable analysts to glean insights from supply chain data. This facilitates decision-making based on evidence rather than intuition. Python helps analysts more correctly forecast demand, estimate delivery schedules, and predict inventory levels. Furthermore, reinforcement learning techniques in Python's machine learning capabilities can enhance vehicle routing, warehouse storage sites, predictive maintenance, and other features. Businesses can get information on inventory levels, transportation costs, demand forecasting, and other topics by utilizing data and analytics. Python is a useful tool for supply chain optimization and analytics. Algorithms utilizing machine learning, for instance, can identify trends and forecast ideal stock levels. [8]

2.2.5 Role of AI in Inventory Management

AI technologies hold the potential to completely transform the way firms manage their inventories by optimizing workflows, reducing expenses, and improving overall effectiveness. It's anticipated that this shift would affect small manufacturers and distributors in addition to big businesses.[9].

Artificial intelligence (AI) systems can forecast future demand by analyzing past sales data. This helps firms avoid stockouts or overstocking by helping them maintain ideal inventory levels. For improved predicting accuracy, AI can also take seasonal patterns, market movements, and other variables into account. AI, for example, can spot a trend in which demand for a given product rises during important athletic events. Then, by keeping an eye on event calendars and historical sales trends, it can forecast these increases in demand. To seize these chances, the business can modify its marketing plans and inventory levels [9].

2.2.6 Role of GPS Tracking in Supply Chain Management

GPS technology allows for real-time tracking of shipments, allowing for ongoing monitoring and location updates of items as they move through the supply chain. GPS improves visibility and control over the whole supply chain by giving exact position data and movement insights. This enhances operational efficiency by enabling businesses to minimize delivery delays, manage shipment routing, and react quickly to any unforeseen events. To guarantee customer pleasure and uphold confidence, real-time tracking gives stakeholders access to reliable information on shipping status. [10]

Items that need to be stored at a given temperature are tracked using GPS tracking devices to keep an eye on their temperature and humidity levels, guaranteeing that they are preserved and meet quality standards all the way through the supply chain. Businesses may reduce the danger of item malfunction and guarantee that goods arrive at their destination in ideal shape by using GPS tracking to precisely monitor the environmental conditions of their items throughout transportation and storage. Because it allows for real-time monitoring and notifications for any deviations from the necessary temperature and humidity ranges, this technology is essential to quality assurance. By offering thorough documentation of the environmental conditions to support adherence to industry norms and laws, it simplifies regulatory compliance. [10]

Synthesis:

The systems and literature reviews highlight how important technology is to supply chain management. Systems like Anaplan, Magaya, and Netstock provide several ways to improve operational effectiveness using integrated solutions. This is by using automation, boosting customer service, and increasing data accuracy for system functionalities.

In this project, the developers will create a system that has similar functionalities to the systems presented above. However, what's new with our system is we will integrate AI for inventory optimization. We also plan to create automation features such as barcode scanning, and automated data entry to minimize human errors and streamline operations.

III. Current Systems

3.1 Current System

Medical hospital equipment distributor MEGAION still does manual inventory checking. The inventory area of the organization uses Google Forms for entering hospital machine equipment descriptions. Following the collection of responses, the staff members compile the data and enter it into the company's Excel spreadsheet.

3.2 Technical Background

MEGAION, a hospital equipment distributor, still uses manual procedures for their inventory checks. The inventory division of the company still uses Google Forms to enter the description of the machine, once the response is gathered, the inventory department staff will finalize the data and put it in excel spreadsheet of the company. Using Google Forms, they indicate whether each item is for demonstration purposes only or for delivery shipment. From there, each equipment status is manually processed by the inventory division of MEGAION.

3.3 List of Processes

Table 1 contains the list of current processes being performed by the client.

Table 1 List of Processes s

Process ID	Process Name	Process Details
P001	**Inquiry**	1. The sales representative identifies and qualifies potential buyers. After identifying a prospect, the sales representative will determine the client's needs and wants.

		2. Clients can also contact the company through their website and fill out the "Request a Quote" section to specify what they need.
P002	**Billing Statement**	1. The procurement staff makes the list of items to be ordered according to the client’s purchase order. 2. Upon receiving a client's Purchase Order from the Sales Department, check if ordered items are in stock or need to be ordered from suppliers. Identify any out-of-stock items requiring purchase. 3. The accounting staff will prepare the sales invoice and statement of account for the client.
P003	**Shipment and Order of Product**	1. The Department Head will make the order with the supplier by email by providing the approved Purchase Order. 2. The Accounting Department will receive an invoice from the supplier. After receiving the invoice, the accounting staff will review it and submit it to the Department Head for payment approval of the product/machine. 3. If the item to be shipped is a radiation device, apply for Clearance for Customs Release (CFCR) online. After submitting the application, pay the provided payment order through Landbank. 4. After compiling the necessary documents for the CFCR application, email them to the FDA Officer. 5. Once paid, the BOC System will transmit an online release order to the broker system. After paying other bills (forwarder), the items will be released and delivered to our warehouse. 6. The Asset Department will be informed of the upcoming delivery, provided a copy of the Packing List, and notified of the delivery date, time, and trucks to request the necessary gate pass.
P004	**Inventory Checking**	1. Once the product is received, the warehouse manager will log the machines to update the inventory. 2. After logging the machine, the inventory manager and the service engineer will handle the product and determine whether it is for demonstration or delivery. 3. Lastly, sorting out the products to determine if the machines need to be placed elsewhere because they are hazardous or not.
P005	**Delivery for Client**	1. Machine for Delivery Assignment. 2. Endorse document to Engineer. 3. Prepare machine for loading. 4. Load & secure machine to vehicle. 5. Driver briefing before travel. 6. Unload and store machines away from hazards.

3.4 Gap Analysis

The Gap Analysis is used to understand MEGAION company's insights into various pain points. It helps identify necessary improvements and outlines how the team can assist in establishing an effective operational system.

It includes SWOT analysis in *Table 2* for better understanding of the strengths, weaknesses, opportunities, and threats of the company. Fishbone diagram in *Figure 1* identifies the focus pain point of the team DHAN INVENTORY system.

3.5 SWOT ANALYSIS

Table 2 SWOT Analysis

SWOT ANALYSIS	
STRENGTH • Market Demand • Customer Relationships • Regulatory Compliance	WEAKNESS • High Operational Costs • Inventory Management • Equipment Technical Failures
OPPORTUNITIES • Marketing and Branding • Market Expansion • Technological Upgrades	THREATS • Supply Chain Disruptions • Cybersecurity threats

This SWOT analysis is about the strength, weakness, opportunities and threats for our client. Our client or their company which is the MEGAION handle’s medical equipment buying, storing and selling.

3.6 FISH BONE DIAGRAM

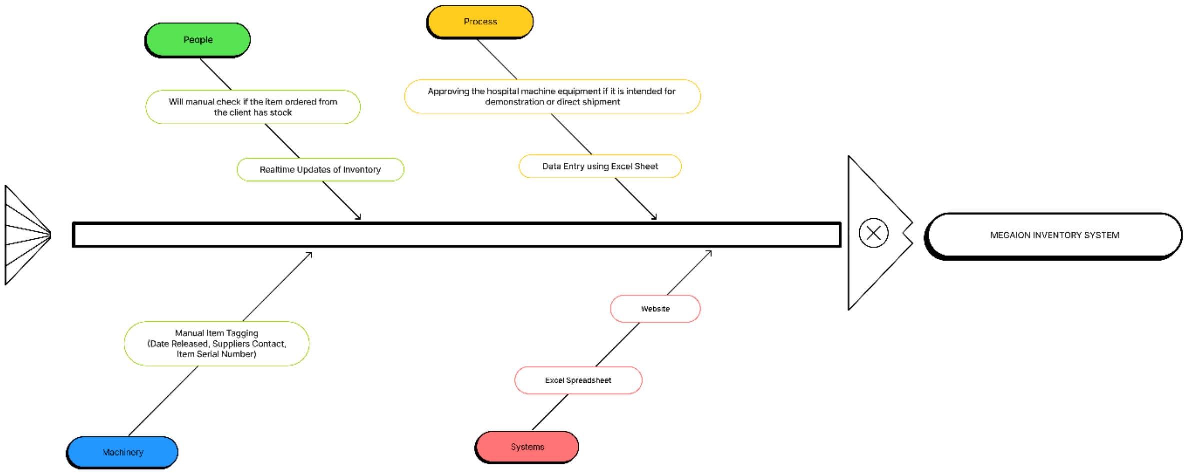


Figure 1

Moving forward, we utilized the Fishbone diagram to gain a clearer understanding of the causes and effects influencing the MEGAION company as shown in the image above.

IV. Proposed Solution

4.2 Lean Canvas

4.2.1 Problem

1. **Manual process of tracking stock levels for delivery or demonstration of machines, leading to potential delays**
2. **To ensure optimal inventory and prevent sales disruptions caused by understocking in a B2B supply chain.**
3. **Outdated and inefficient process of machine retrieval after demonstration.**

4.2.2 Solution

1. To reduce the delay of checking stock levels for items that are for delivery or demonstration by 85%.

2. To ensure optimal inventory and prevent sales disruptions caused by understocking.

3. To streamline the process for retrieving and logging demonstration machines to mitigate instances of missing stock.

4.2.3 Key Metrics

1. *Customer Satisfaction*
2. *Customer Retention*
3. *Return of Investment*

4.2.4 Unique Value Proposition

1. *AI forecasting and automated data cleaning*
2. *Advanced Parts Traceability*
3. *Minimized Human Error*
4. *Visual Representation of Sales*
5. *Downloadable Reports*
6. *Adaptation to all devices*

4.2.5 Customer Segment

1. *Hospitals*
2. *Health Care Organizations*
3. *Educational Institutions*

4.2.6 Channels

1. *Conferencing Platforms (Google Meet)*
2. *E-mail*
3. *Face to Face Interviews*

4.2.7 Revenue Streams

1. *One time Payment*
2. *Support and Maintenance of Application*

4.2.8 Cost Structure

1. *Hosting Costs*
2. *Barcode Scanners*
3. *Sensors*
4. *Development Costs*

4.2.9 Unfair Advantage

1. *Customization and Flexibility*
2. *Data Analytics*

4.3 Product Vision

To innovate the efficiency and accuracy of the megaion medical equipment management operations by developing an integrated inventory management and supply chain management system equipped with a barcode management solution that has ai. This will enable real-time tracking of inventory, streamline supply processes, time efficient, and improve overall customer satisfaction.

Table 3 Product Vision

For: Megaion Inventory and logstics
Who: Has the responsibility for monitoring the inventory and determining whether the equipment is for delivery, demo, or just inventory
The: DHAN System
That: Monitors the inventory, has a supply chain management system, a barcode scanning and ai feature
Unlike: The traditional method for them to use pen and paper and google sheets before putting it in excel.
Our: Inventory system will be more efficient and less time consuming for the user

4.4 Technology Specifications

The project will be used as a tool that will only focus on the client's needs. Moreover, the team proposed to create an inventory system that will solve the client's problems. This system will use barcode scanning to see if an item is for demonstration or for delivery. For now, the developers plan to develop the project using HTML, CSS, JS, SQL, PHP, PYTHON, Visual studio, Pandas Python and AdminLTE. Initially, the project will be developed to address the client's current needs. In future development, more features would be added such as AI, invoice printing, order management screen, and improved notification systems. Lastly, the system will have a customizable dashboard wherein you can access data insights that can be exported into detailed reports.

Pandas is a Python library designed for data manipulation and analysis. It offers fast and flexible data structures like Series and DataFrame, similar to tables in databases, for handling large datasets. Pandas supports operations such as data alignment, handling missing data, and reshaping datasets. Accurate demand forecasting is crucial for optimizing inventory levels in supply chain management, preventing overstock or stockout situations. By analyzing historical data, industry trends, and consumer behavior, organizations can enhance forecasting accuracy. Python provides libraries for statistical operations, data visualization, and predictive modeling, aiding in informed decision-making for inventory management.

The system will also make use of PubNub for real-time item tracking. In order to deliver current information on the location and status of things, PubNub provides real-time messaging and data streaming features. By ensuring that all parties involved have access to the most recent information, better decision-making and more effective operations are made possible.

The following is the category and specifications needed to develop the project:

Table 4 List of Category and Specifications

Category	Specifications
Hardware	• Barcode Scanners • Computer/Laptop • Router
Software	• MySQL • AdminLTE • Visual studio • Python Panda • PubNub • JavaScript
Peopleware	• Project Members • Project Adviser • Project Consultant • Client
Network	• Wide Area Network (WAN)

4.5 Feasibility

4.5.1 Technical Feasibility

The project will be implemented with tools that can increase the system's efficiency. The project’s compiler will be VS code with JavaScript and PHP as it’s main language. With JavaScript, there would be a lot of libraries that can be used e.g (jQuery, MorrisJS, ChartJS) to increase the functionality of the system. AdminLTE would also be used because it is made for creating admin panels, dashboards, and back-end user interfaces. All of the tools given would be used to develop the system.

4.5.2 Market Feasibility

The client’s main problem, which is the monitoring of items that goes in and out of the inventory, it shows that inefficiency and human errors are the main concern. With this project, the efficiency of handling the inventory will surely increase and will reduce human errors. The marketable feasibility for the project is to greatly improve the handling of inventory, thus helping employees save time and reduce potential errors. Employees can now focus on advertising their products and accommodating more customers.

4.5.3 Operational Feasibility

The client and its employees fully support and believe that this project will solve their current issues in inventory operations. According to the client, there are times they encounter conflicts in their inventory due to unorganized excel sheets. This project will help the client organize their needs by simply downloading excel sheets from sales reports provided by the system. Furthermore, the developers will provide a tutorial to the employees on how to use the system. The group will be open to any suggestions the client gives to improve the user-friendliness of the said system.

V. Requirements Analysis

5.1 Product Backlog / User Stories

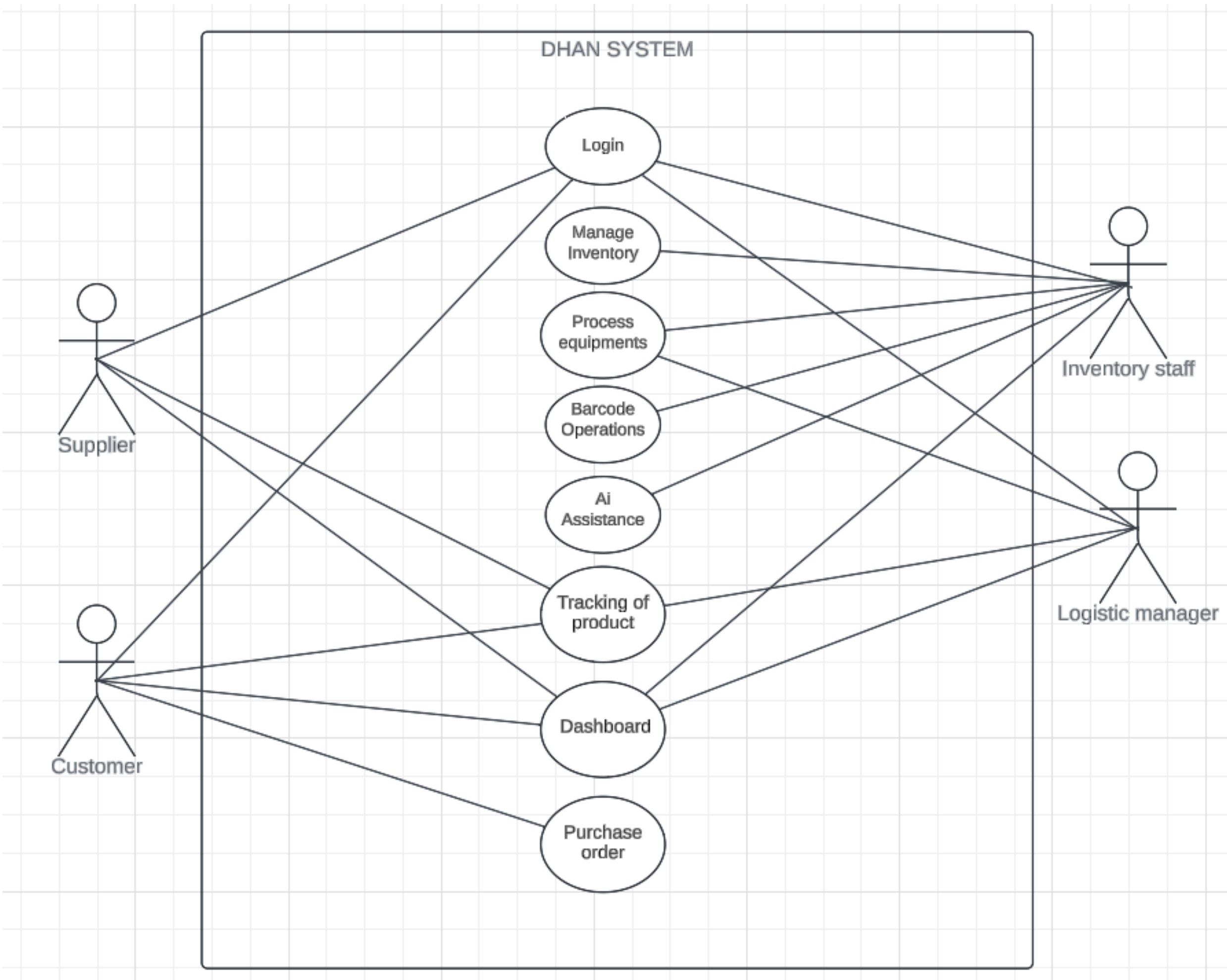
Table 5 Product Backlog

DHAN INVETORY SYSTEM					
ID	As a..	I want to be able to..	So that..	Priority	Status
1	Inventory staff	Scan equipment’s using barcodes	I can quickly add them to the inventory system	Must	Not yet started
2	Inventory staff	Use AI to forecast	I can maintain optimal stock levels	Must	Not yet started

		inventory restock needs			
3	Inventory staff	Generate barcodes	Each equipment is uniquely identifiable	Must	Not yet started
4	Inventory staff	Receive low stock alerts	I can order items before they run out	Must	Not yet started
5	Inventory staff	Track the equipment's status	I know if the equipment is for delivery/demo/stock/rent/borrowed	Should	Not yet started
6	Inventory staff	Access a dashboard for inventory supplies	I can monitor the equipment's status in real-time	Must	Not yet started
7	Inventory staff	Use AI for predictive analytics	I can optimize inventory based on the equipment stocks	Must	Not yet started
8	Inventory staff	Receive notification via email	I can know if we are on low stocks on a certain equipment	Must	Not yet started
9	Inventory staff	Use AI assistance	It can help the staff to predict stock shortages and optimize inventory levels	Must	Not yet started

5.2 Use Case Diagram

Figure 2 Use Case Diagram



5.3 User Classes and Characteristics

Table 6 User Classes and Characteristics

Roles	Description
Logistics manager	The logistics manager is responsible for tracking the equipment's status and whereabouts.
Inventory Staff	This user has access to adding and deleting items for the inventory. They will also be the one to scan machine barcodes that would determine if a product is for demonstration or for delivery.
Supplier	The supplier can login, track shipments and can update the stock if it’s confirmed if the shipment is complete via shipment tracking and notification.
Customer	The customer can login in and can place their order. If its completed they can monitor it via tracking of the product and have a notification canceled or completed.

5.4 Prototype

Below is Figure *3 of* the prototype login page. Only designated users of the system will be able to access it.

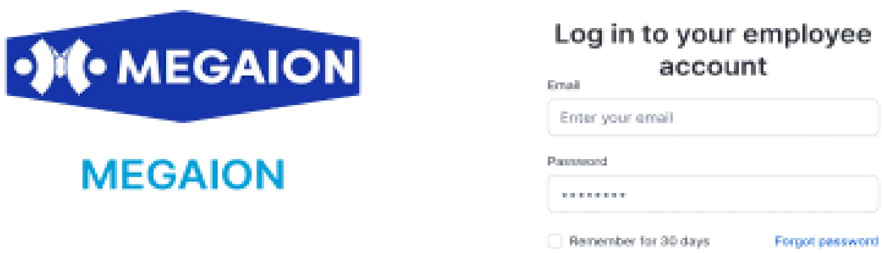


Figure 3 Login Page

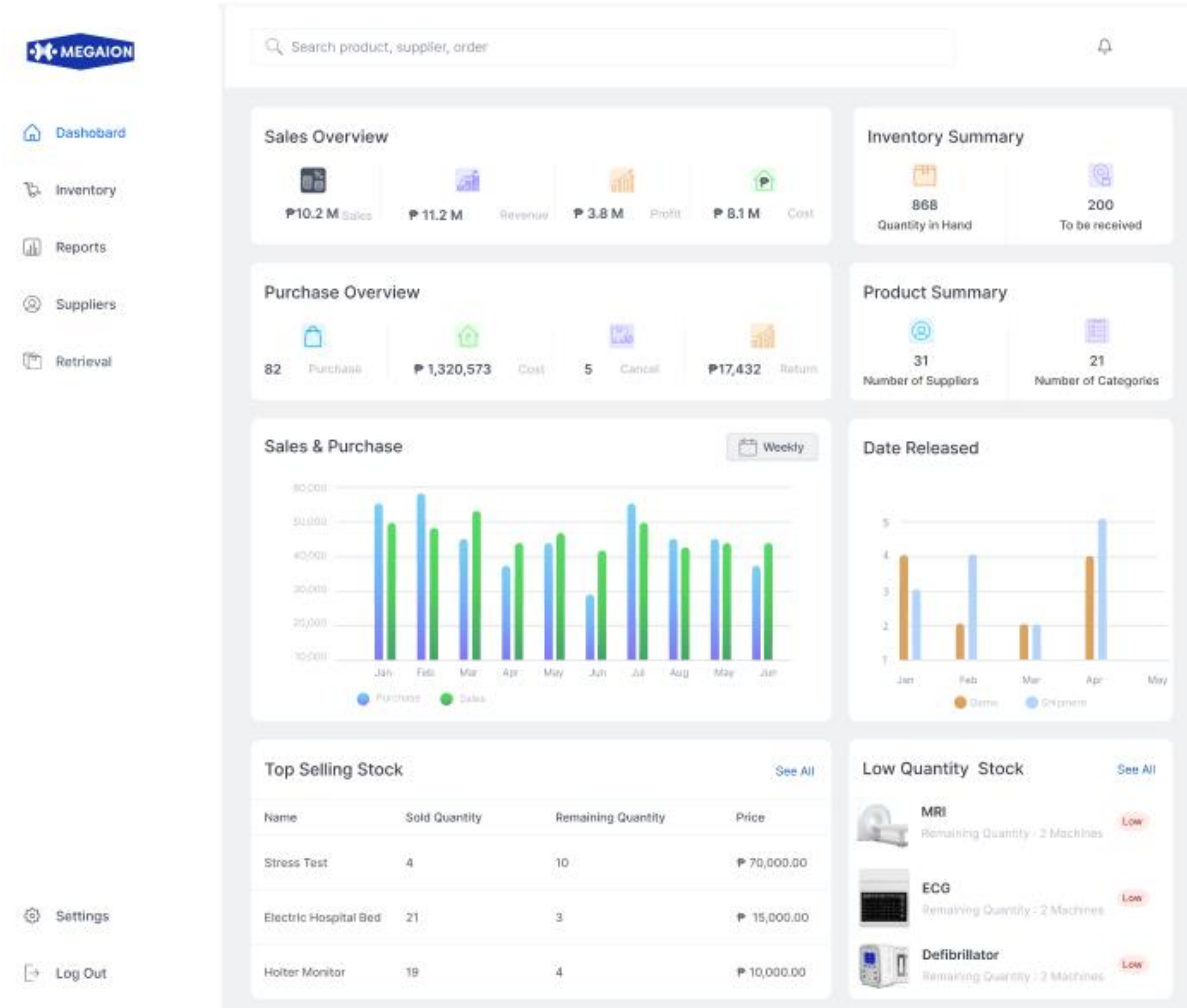


Figure 4 Landing Page

In *Figure 4,* The company's landing page leads to the Inventory System dashboard, where the users may view the sales overview, purchase overview, inventory summary, and product summary. Moreover, the dashboard will also show the top selling product and its date of release. Lastly, users would also know what products are low in quantity.

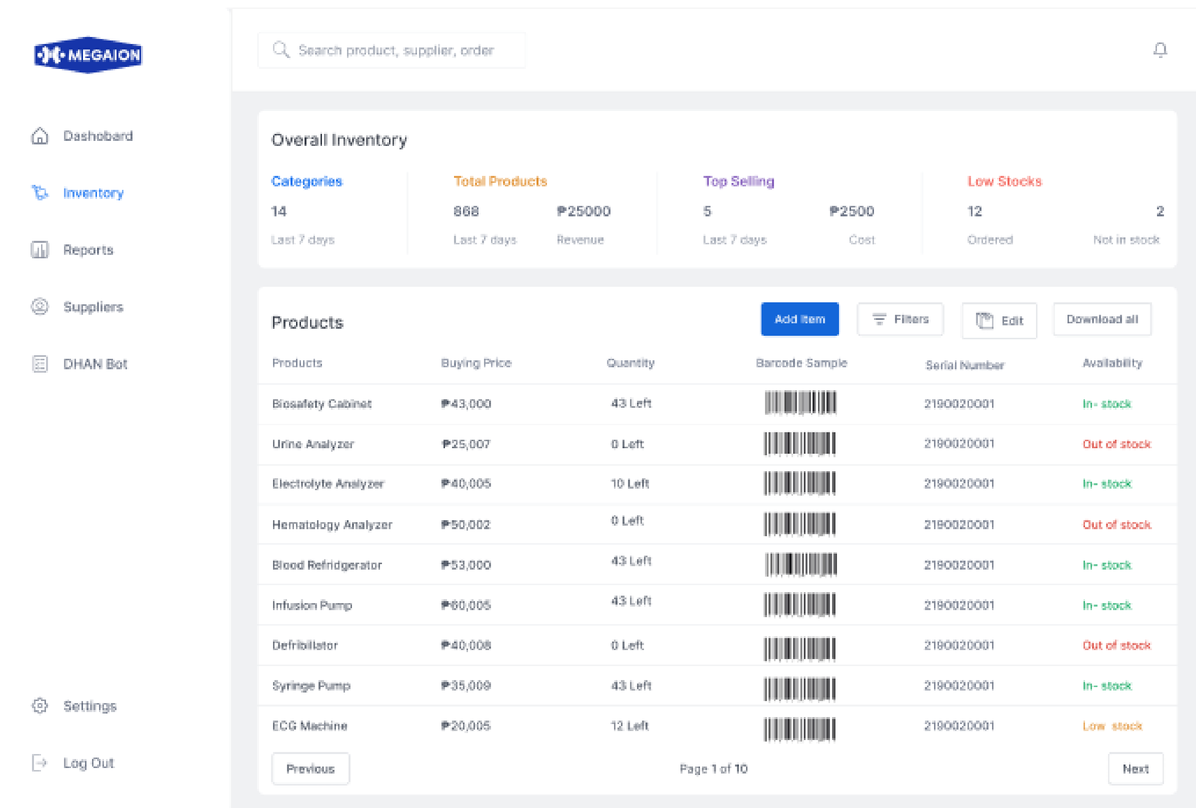


Figure 5 Inventory Tab

In Figure 5, this tab of the dashboard shows the overall products and the state of every equipment unit. It also shows the stock level status. After each product is entered, a unique barcode is automatically produced.

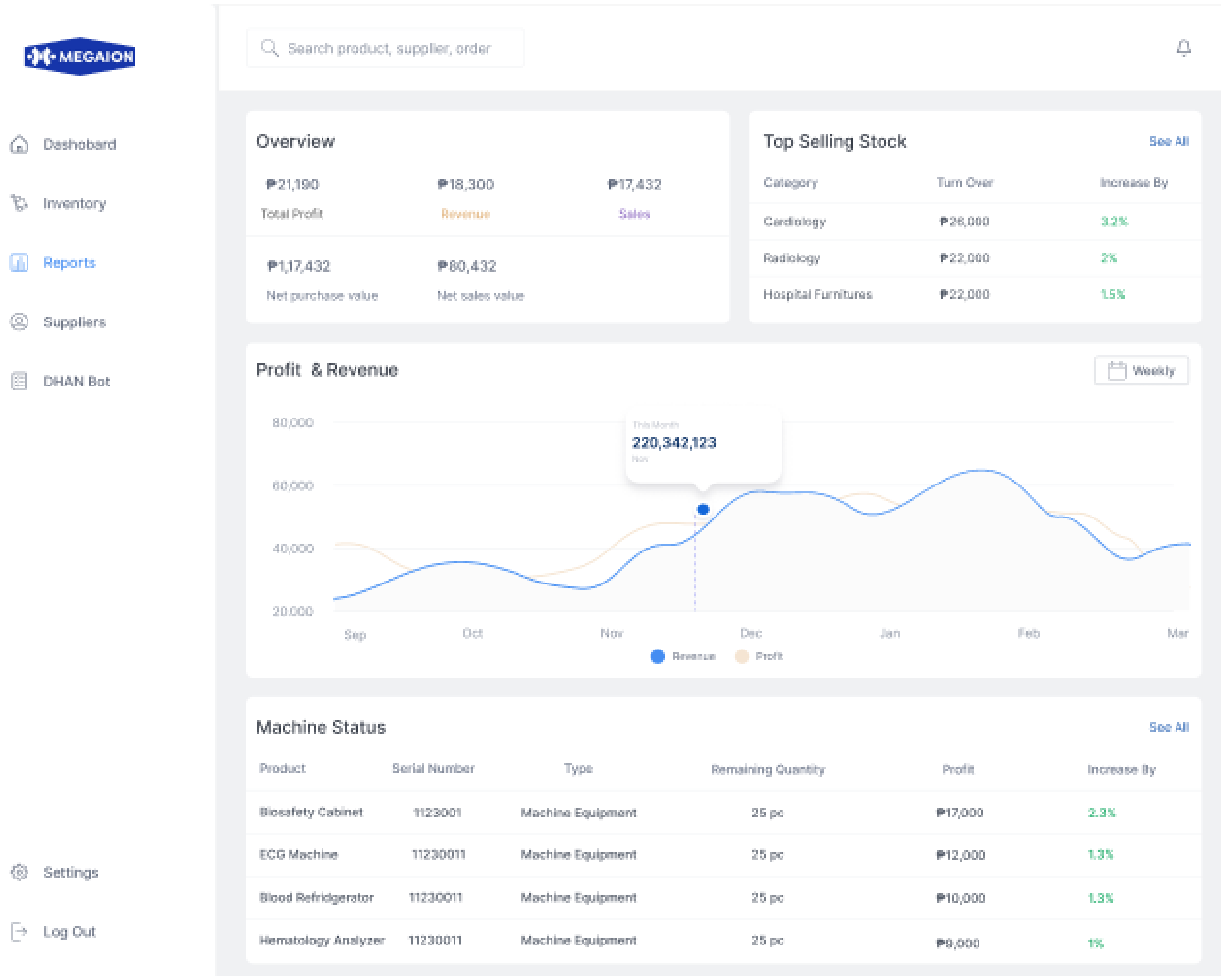


Figure 6 Reports Tab

In Figure 6, the use of predictive analytics in reports helps provide insights into potential future sales patterns.

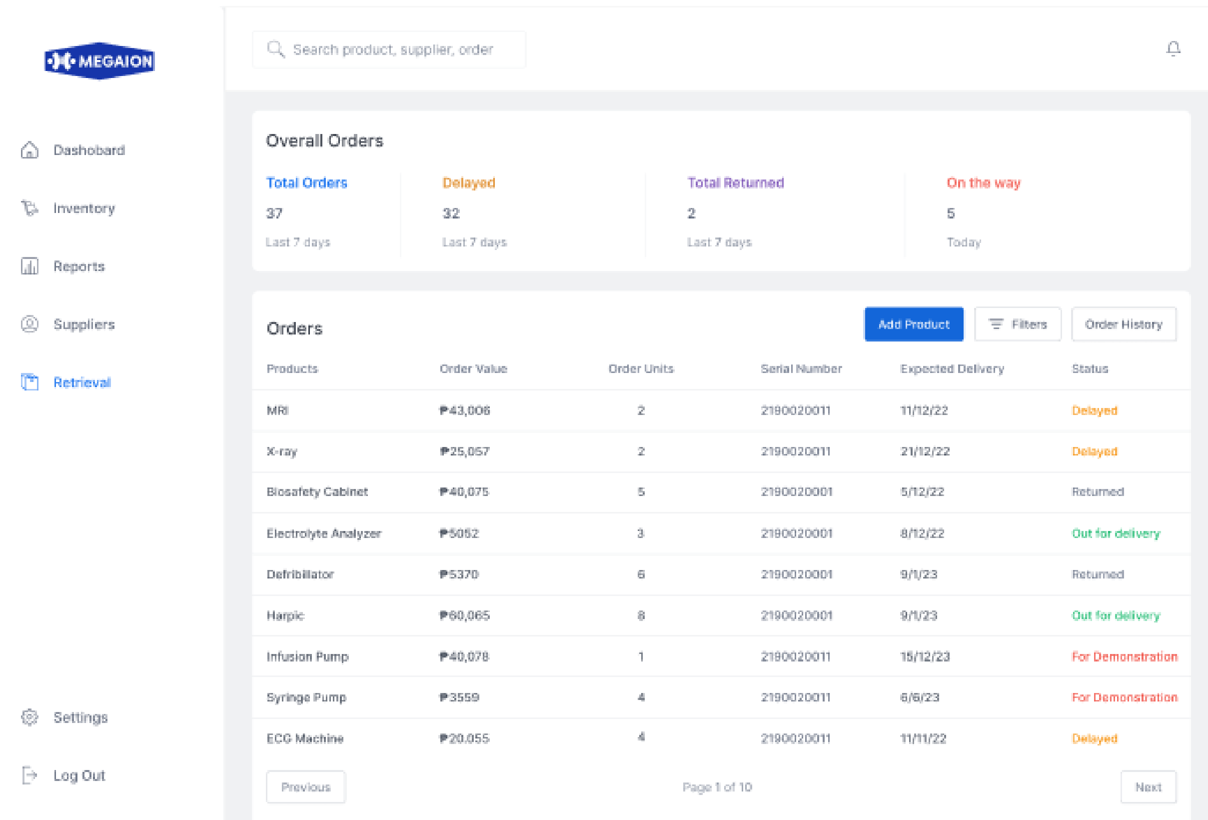


Figure 7 Shipment Status

In Figure 7, this section will display the shipment status of each in and out product. This allows the assigned staff of the inventory resource department to track the shipping status of products intended for clients.

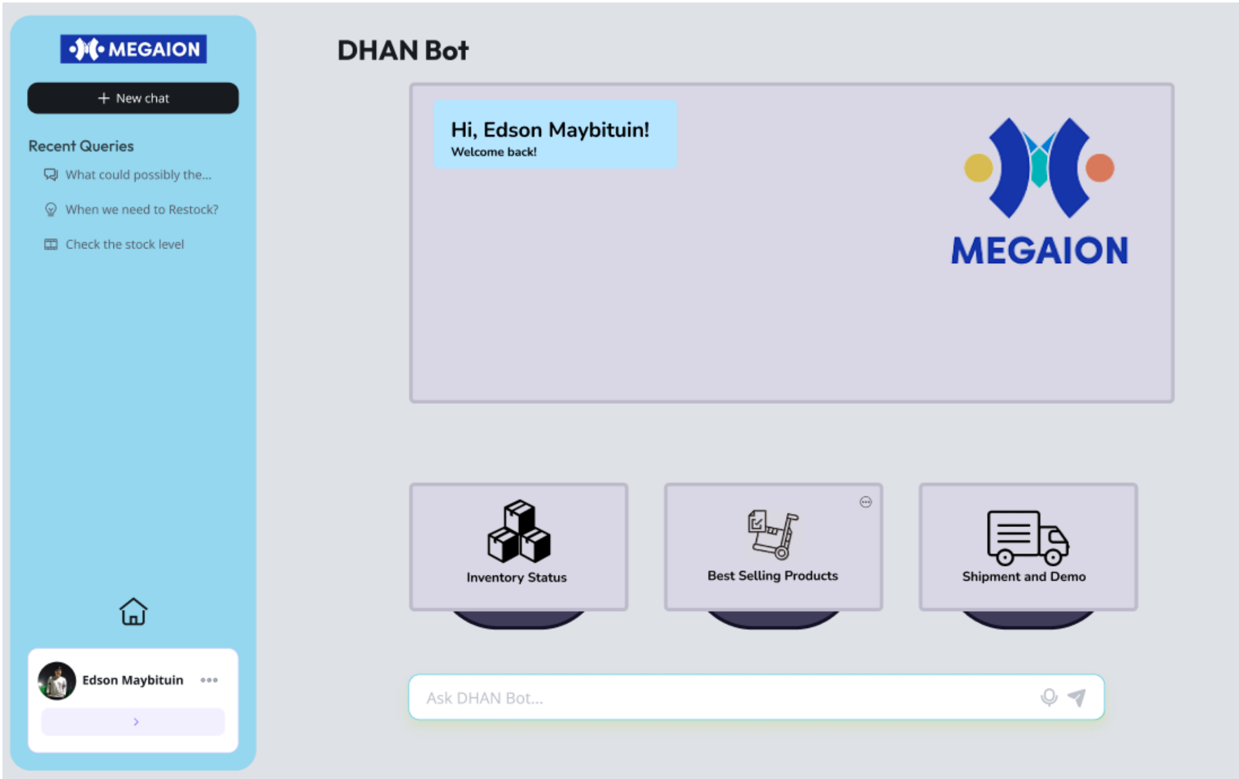


Figure 8 AI Module

In figure 8, this section will display the prototype AI module dashboard. The AI can help users track product demand, stock levels, and see graphs.

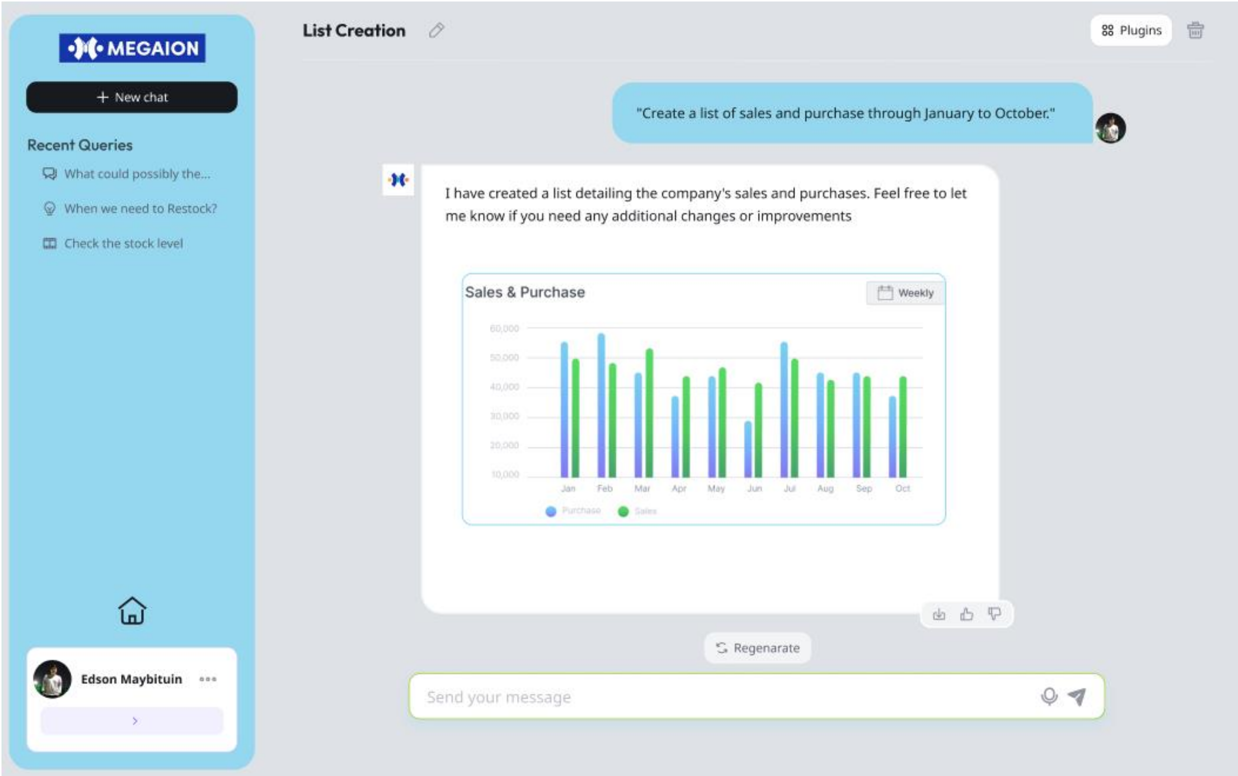


Figure 9 AI Module

In figure 9, this section will an example of an AI query that displays the company's sales and purchases. The AI also provided a bar graph for users to know what they asked thoroughly.

5.7 Release Plan

Target Group: Megaion (Inventory department employees)

Goal: To improve operational system by implementing an advanced inventory system that integrates a supply chain management system, barcode system and AI capabilities. This system will streamline the process of tagging and tracking medical equipment, ensuring accurate and real time management of their inventory

Needs: A software solution that optimizes the working efficiency of the staff and assists in completing activities efficiently.

Value: To have a more accurate inventory management, improve the accuracy in tracking of their medical equipment's and over all enhance their work efficiency

Key features:

- User log in
- Real-Time inventory updates
- Supply Chain Management System Integration
- AI assistance
- Barcode system
- Notification system
- Dashboard
- Product tracking

Release 1:

- Project documentation
- Creating project blueprint
- Project prototype design

Release 2:

- Database modeling
- Implement of database system planning and setup
- Developing wireframe and UI/UX design
- Research and design overall systems

Release 3:

- Implementation of Supply Chain Management System, AI module, and barcode systems with inventory management
- System testing
- Fix any bugs or defects
- Project deployment

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Appendices

Appendix A: Project Vision

Our mission is to create an automated inventory management with a supply chain management system that will transform the accuracy and reduction of delay of MEGAION medical equipment warehouse operations. With the help of artificial intelligence (AI) and a sophisticated barcode management system, this system will be able to track inventories in real-time, streamline supply reports, and ultimately improve work efficiency.

This project began with client interviews and problem identification. This was followed by proposing a solution that will solve the client's pain points. After proposing a solution, a brainstorming was made to think about what other features can be implemented in our system. The group made sure that the proposed features were inline with the client's needs.

Appendix B: Schedule/Release Plan

Figure 10 Schedule Plan

This project schedule plan will show our timeline to divide our task into three phases MNSTDEV, MSYADD1 and MSCPROJ making it one year from April 2024 to April 2025

Appendix C: Product Roadmap

(

Updated product roadmap file here. Make sure that content is readable

)

Table 7 Product Road Map

MNSTDEV	MSYADD1	MCSPROJ
Milestone 1 • Looking for a client • Brainstorming and delegating task on MS teams and messenger **Milestone 2** • Meeting and interviewing the client • Making the research paper • Creating the design UI/UX design via Figma **Milestone 3** • Consulting our adviser and consultant about the paper • Presentation of the paper to our adviser and panelist • Finals defense	**Milestone 4** • Learn and adjust the paper and project due to panelists advise **Milestone 5** • Research and design the overall systems of our project • Database modeling	**Milestone 6** • Creating the system prototype • Implementation of AI module and barcode systems with inventory management **Milestone 7** • Testing of the system and bug hunting and fixes • Project deployment

Appendix E: Teams Meetings

(

Include here screenshots of meetings with adviser/consultants (please ask permission), and if possible, minutes of your meetings

Date

Agenda

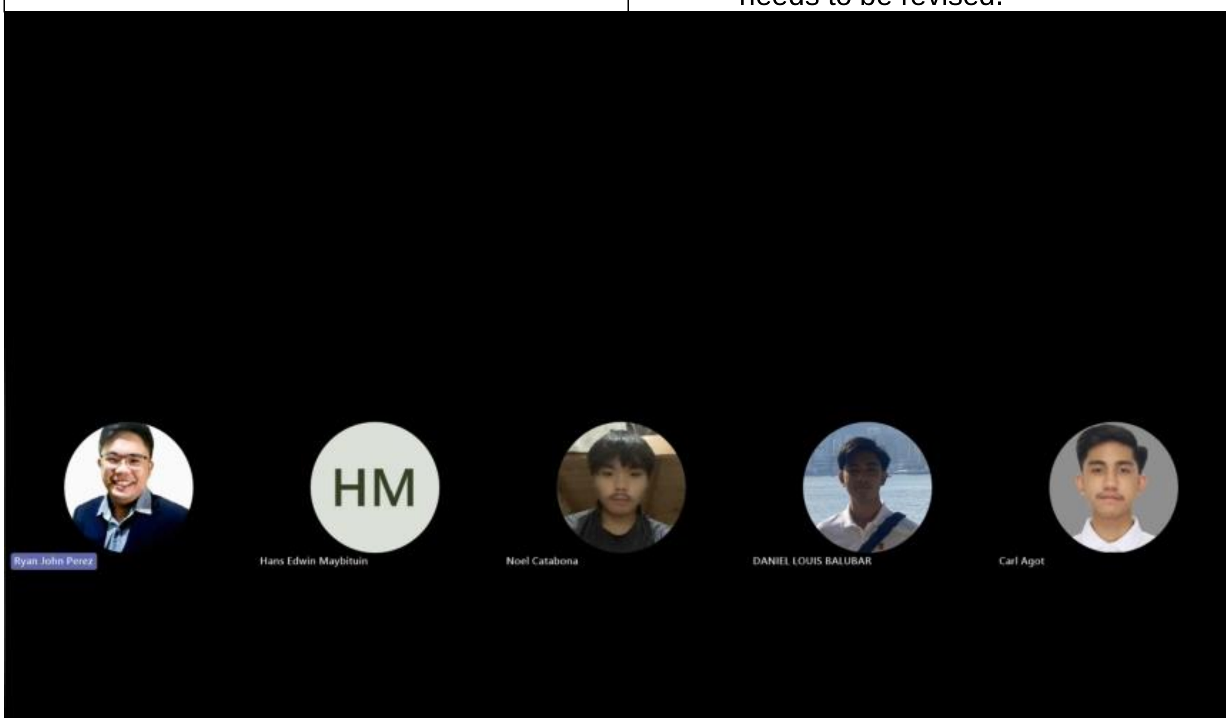
Screenshot

)

Date: June 10, 2024

Time: 4:24 PM – 5:20 PM

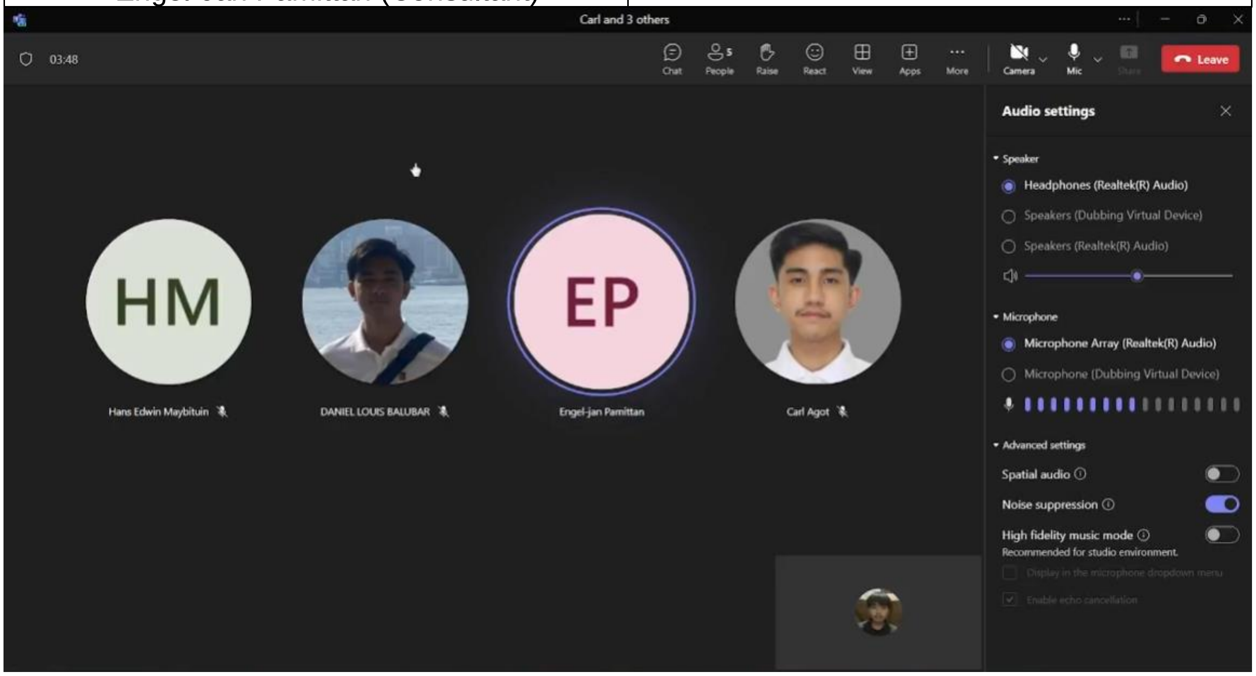
Attendees	Purpose / Objectives
• Hans Edwin Y. Maybituin • Daniel Louis Balubar • Carl Agot • Noel Catabona • Ryan John Perez (Adviser)	• Consultation about the post midterm presentation • Discuss the strengths and weaknesses of the post midterm presentations. • Highlight areas that met or exceeded expectations, and highlight also what needs to be revised.



Date: June 14, 2024

Time: 1:30 PM – 3:30 PM

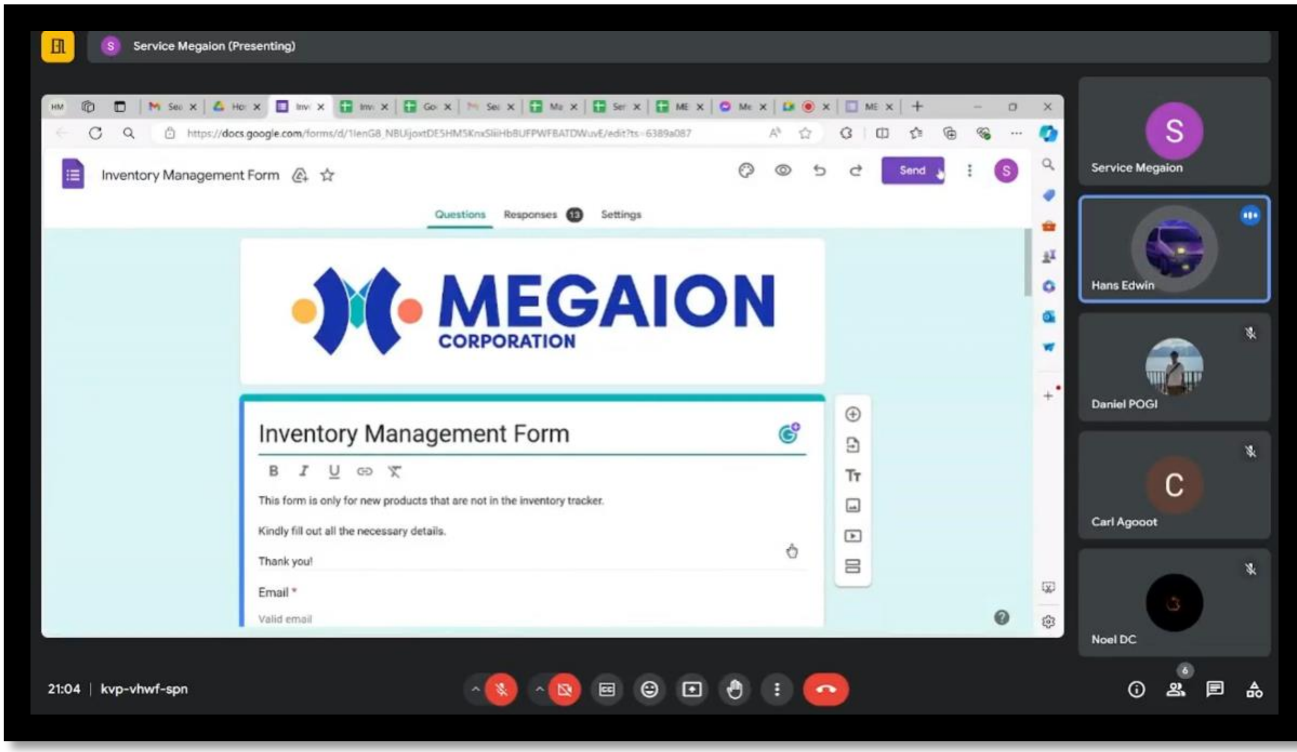
Attendees	Purpose / Objectives
• Hans Edwin Y. Maybituin • Daniel Louis Balubar • Carl Agot • Noel Catabona • Engel-Jan Pamittan (Consultant)	• Revisions of Paper. • Our consultant suggested a pos system to develop the inventory system



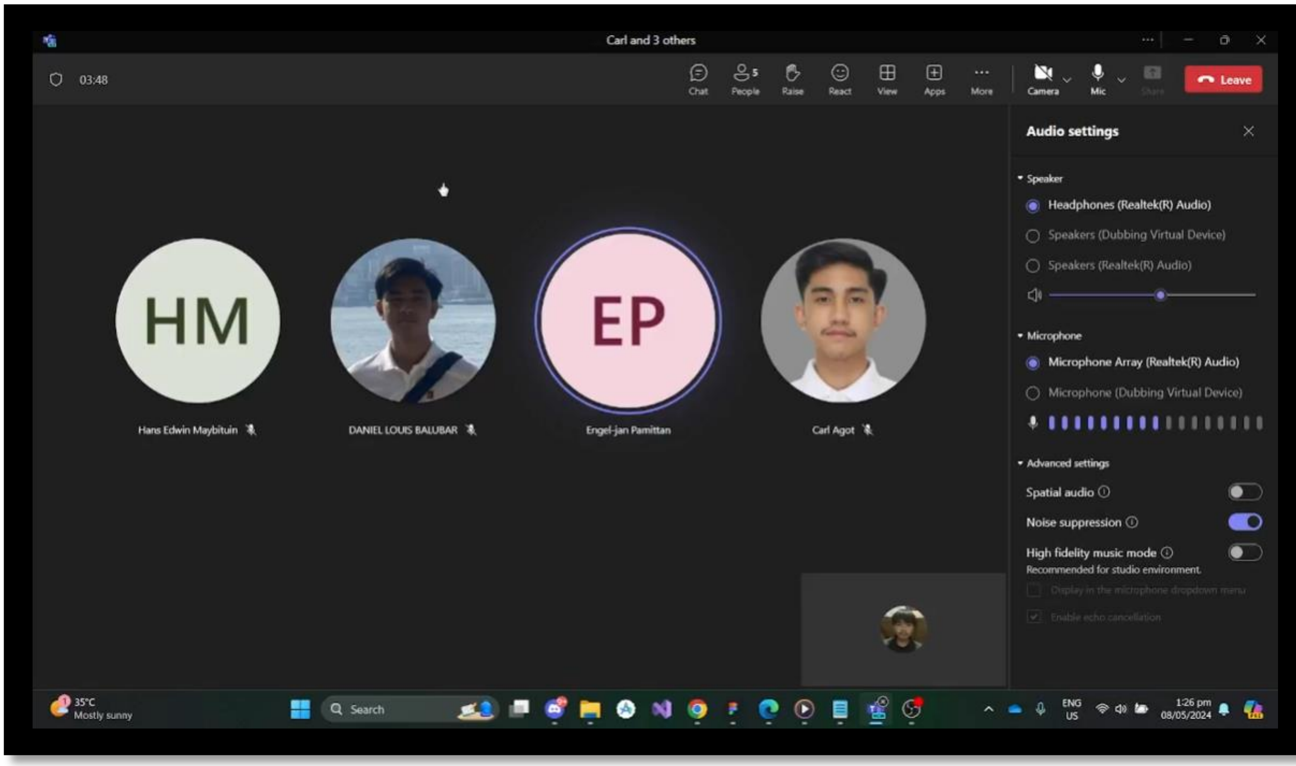
Date: June 24, 2024

Time: 7:00 PM – 8:00 PM

Attendees	Purpose / Objectives
• Hans Edwin Y. Maybituin • Daniel Louis Balubar • Carl Agot • Noel Catabona • Edson Y. Maybituin	• Final Meeting with the Client. • Compilation of the Project and the requirements for Presentation. • Showing the Client the Figma/Framework of our Solution System.



Service Megaion (Presenting)
Inventory Management Form
Send
Questions
Responses
Settings
MEGAION
CORPORATION
Inventory Management Form
This form is only for new products that are not in the inventory tracker.
Kindly fill out all the necessary details.
Thank you!
Email *
Valid email
Service Megaion
Hans Edwin
Daniel POGI
Carl Agooot
Noel DC
21:04 | kvp-vhwf-spn


Carl and 3 others
03:48
Chat
People
Raise
React
View
Apps
More
Camera
Mic
Leave
Audio settings
Speaker
Headphones (Realtek(R) Audio)
Speakers (Dubbing Virtual Device)
Speakers (Realtek(R) Audio)
Microphone
Microphone Array (Realtek(R) Audio)
Microphone (Dubbing Virtual Device)
Advanced settings
Spatial audio
Noise suppression
High fidelity music mode
Recommended for studio environment.
HM
EP
Hans Edwin Maybituin
DANIEL LOUIS BALUBAR
Engel-jan Pamittan
Carl Agot
35°C
Mostly sunny
Search
ENG
US
1:26 pm
08/05/2024